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## Original Article

## Prognostic factors and survival outcomes in breast cancer patients with brain metastases: A retrospective study

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## ABSTRACT

**Background:** Brain metastases are a significant complication of breast cancer and associated with low survival rates. However, the prognostic factors that influence outcomes after the diagnosis of brain metastases are not well-defined, especially for patients in Thailand. This study aims to identify the prognostic factors linked to overall survival in breast cancer patients who develop brain metastases.

**Material and methods:** This study included 40 patients with breast cancer and brain metastases. Overall survival from brain metastasis diagnosis to death or last follow-up was analysed using Kaplan–Meier methods, with comparisons by log-rank test. Univariate and multivariate Cox regression analyses were performed to identify prognostic factors.

**Results:** The median OS for the cohort was 4.0 months (95% confidence interval: 3.23–4.77), with survival rates at 6 and 12 months being 27.5% and 17.5%, respectively. Patients older than 60 years had a worse survival rate compared to those under 50. While hormone receptor-positive patients tended to have slightly better OS than those with HER2-positive and triple-negative subtypes, breast cancer subtype did not significantly affect survival outcomes. Patients who underwent surgery combined with whole-brain radiotherapy (WBRT) achieved the best results. In the multivariate analysis, age over 60 was identified as an independent predictor of poorer OS, with a hazard ratio of 4.38 (95% CI: 1.06–18.11;  $p = 0.041$ ).

**Conclusions:** Survival rates following the diagnosis of brain metastases in breast cancer patients remain low. Advanced age serves as an independent adverse prognostic factor, while aggressive treatments targeting the brain have been linked to improved survival outcomes.

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## 1. Introduction

Breast cancer is the most prevalent cancer among women globally, and its incidence rates are continually increasing. In Thailand, these rates have risen from 20.5 to 35.7 cases per 100,000 person-years over the last two decades. Additionally, breast cancer is the second leading cause of brain metastases, following lung cancer, with 20% to 40% of breast cancer patients ultimately developing these secondary tumors.<sup>1</sup> Current treatment options include whole brain radiation therapy (WBRT), surgical procedures, and stereotactic radiosurgery (SRS). Nonetheless, the presence of the blood-brain barrier limits the effectiveness of

systemic therapies, resulting in a challenging prognosis, with median overall survival rates ranging from 4.5 to 13.8 months for these patients<sup>2–4</sup>.

Brain metastases are more common in HER2-positive (HER2+) and triple-negative breast cancer (TNBC) subtypes<sup>5,6</sup>, but data on their incidence in Thailand are lacking. Given their global prevalence, similar trends are likely in the Thai population. Although hormone receptor-positive (HR+) breast cancer typically has a lower risk of brain metastasis, increasing endocrine resistance has been linked to a rising incidence.<sup>7,8</sup> In contrast, HER2+ and TNBC subtypes have a higher potential for metastases, with poor outcomes often due to limited blood-brain barrier penetration and a lack of targeted therapies.<sup>9,10</sup> Management of brain metastases in breast cancer is complex and typically involves a multimodal

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E-mail address: [ghent\\_911@hotmail.com](mailto:ghent_911@hotmail.com) (P. Kongon).<https://doi.org/10.1016/j.asjsur.2026.03.194>1015-9584/© 2026 Asian Surgical Association and Taiwan Society of Coloproctology. Publishing services by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

approach. Despite advances in systemic and local therapies, overall survival in patients with breast cancer and brain metastases remains poor and may be influenced by factors such as molecular subtype, brain-directed treatment modality, and visceral metastatic burden. Understanding prognostic factors specific to breast cancer patients with brain metastases is essential for optimizing treatment decision-making, counselling patients and families, and allocating healthcare resources appropriately. Identifying independent predictors of survival may also help refine risk stratification and guide individualized treatment approaches.

Therefore, this study aimed to identify prognostic factors for overall survival in breast cancer patients with brain metastases by evaluating clinical characteristics, breast cancer subtype, extracranial disease, pattern of brain metastases, and treatment modalities using survival analysis.

## 2. Methods

### 2.1. Study design and population

This retrospective cohort study was conducted at the Department of Surgery at Panyanaphikkhu Chonprathan Medical Centre. We reviewed the medical records of patients with histologically confirmed breast cancer who developed brain metastases and received treatment at our institution from March 2010 to March 2025. Eligible patients were aged 18 to 75 years and had a confirmed diagnosis of brain metastases based on radiologic imaging. Patients were excluded from the study if their clinical or pathological data were incomplete, and those who were lost to follow-up were also excluded from the analysis.

### 2.2. Data collection

We reviewed the medical records of patients and collected their clinicopathological data, including age, tumor size, histological grade, histological subtype, extracranial metastases, and the status of estrogen receptors (ER), progesterone receptors (PR), and human epidermal growth factor receptor 2 (HER2). We recorded the time to diagnosis of brain metastases (TTBM), defined as the period from the histological diagnosis of breast cancer to the detection of brain metastases, and the treatment modality for brain metastases.

### 2.3. Statistical analysis

Continuous variables were summarized as mean  $\pm$  standard deviation (SD) or median with interquartile range (IQR), depending on their distribution. Categorical variables were presented as frequencies and percentages. Demographic and clinical characteristics between patients with HR + breast cancer and those with HER2+ or TNBC were compared using independent t-tests for continuous variables with normal distribution, and chi-square or Fisher's exact tests for categorical variables.

Overall survival (OS) was defined as the time from the diagnosis of brain metastases to death or last follow-up, with survival curves estimated using the Kaplan–Meier method and compared using the log-rank test. Univariate Cox regression analysed potential prognostic factors, including age at diagnosis, breast cancer subtype, and treatment modality, with significant variables entered into a multivariate model to identify independent factors. Hazard ratios (HRs) with 95% confidence intervals (CIs) were reported.

All statistical tests were two-sided, with  $p < 0.05$  considered significant, and analyses were conducted using SPSS version 21.

## 3. Results

### 3.1. Patient characteristics

Between March 2010 and March 2025, a total of 270 breast cancer patients were treated at the Department of Surgery, Panyanaphikkhu Chonprathan Medical Centre. Among these patients, 54 (18%) were diagnosed with brain metastases, although 14 were excluded from the study due to incomplete clinical and pathological data. Consequently, 40 patients were ultimately enrolled in the study. The cohort had a mean age of 59.73 years ( $\pm 12.66$ ), with 42.5% older than 60 years, 30% aged between 50 and 60, and 27.5% under the age of 50. The average body mass index (BMI) was 22.54 kg/m<sup>2</sup> ( $\pm 3.61$ ), and 47.5% of the participants were classified as overweight, with a BMI exceeding 23 kg/m<sup>2</sup>. On average, it took 22.55 months ( $\pm 9.10$ ) for these participants to develop brain metastases following their initial cancer diagnosis. Regarding breast cancer subtypes, triple-negative breast cancer (TNBC) was the most prevalent, affecting 45% of patients, followed by hormone receptor-positive (HR+) cancer at 35% and HER2-positive (HER2+) cancer at 27.5%. The mean Ki-67 index was 59.15 ( $\pm 16.88$ ), with the vast majority of tumors classified as grade 3 (92.5%). In terms of lymph node involvement, the N2 stage was the most frequently observed, impacting 32.5% of patients. Additionally, 70% of participants presented with extracranial metastases.

In the management of primary breast cancer, 77.5% of patients underwent surgical intervention, while 72.5% received chemotherapy, with anthracycline and taxane-based regimens being the most commonly utilized (72.5%). Anti-HER2 therapy using trastuzumab was administered to 22.5% of patients. Among those with brain metastases, 62.5% presented with predominantly solitary lesions. The most frequent brain-directed treatment was WBRT alone, accounting for 17.5%, followed by a combination of surgery and WBRT, which represented 12.5%. Furthermore, 65% of patients did not receive any form of local treatment for their brain metastases.

The six-month mortality rate after the diagnosis of brain metastases was 70.7%. The mean overall survival was  $7.0 \pm 6.11$  months, with a median survival of 4.0 months (IQR, 3–11.5 months), as shown in Table 1.

### 3.2. Comparison between HR+ and HER2+/TNBC subtypes

Patients were classified into two groups according to breast cancer subtype: hormone receptor-positive (HR+,  $n = 14$ ) and HER2-positive or triple-negative breast cancer (HER2+/TNBC,  $n = 26$ ) (see Table 2). The baseline demographic characteristics of the two groups were comparable. No significant differences were observed in age distribution ( $p = 0.770$ ) or body mass index (BMI) ( $p = 0.753$ ).

The time from initial breast cancer diagnosis to the development of brain metastases (TTBM) was significantly longer in the HR+ group than in the HER2+/TNBC group ( $27.29 \pm 10.07$  months vs.  $18.65 \pm 6.08$  months,  $p = 0.011$ ), indicating that patients with HR+ disease developed brain metastases at a later stage. In contrast, the HER2+/TNBC group demonstrated significantly higher proliferative activity, as reflected by a higher mean Ki-67 index ( $63.46 \pm 17.19$  vs.  $51.14 \pm 13.44$ ,  $p = 0.026$ ), consistent with the more aggressive biological behavior of these subtypes.

In the HR+ group, all patients underwent surgical resection of the primary tumor, while only 65.4% of the HER2+/TNBC group had surgery (100% vs. 65.4%,  $p = 0.016$ ). This difference is due to the aggressive disease typically seen in HER2+/TNBC patients, making surgery less viable. There were no significant differences in

**Table 1**

Demographic, clinicopathological, and treatment characteristics of breast cancer patients with brain metastases.

| Variables                                      | Count (n = 40)    | Percentage (%) |
|------------------------------------------------|-------------------|----------------|
| Age (years)                                    |                   |                |
| mean $\pm$ SD                                  | 59.73 $\pm$ 12.66 |                |
| median (IQR)                                   | 59 (48–69)        |                |
| 50 or below                                    | 11                | 27.5           |
| 51–60                                          | 12                | 30.0           |
| Above 60                                       | 17                | 42.5           |
| Body mass index (BMI) (kg/m <sup>2</sup> )     |                   |                |
| mean $\pm$ SD                                  | 22.54 $\pm$ 3.61  |                |
| median (IQR)                                   | 22.6 (20.8–24.3)  |                |
| 18.5 or below                                  | 5                 | 12.5           |
| 18.5–22.9                                      | 16                | 40             |
| Above 23.0                                     | 19                | 47.5           |
| Time to diagnosis of brain metastases (months) |                   |                |
| mean $\pm$ SD                                  | 22.55 $\pm$ 9.10  |                |
| median (IQR)                                   | 20 (15–28)        |                |
| Histologic subtypes                            |                   |                |
| Hormone receptor positive                      | 14                | 35.0           |
| HER2- positive                                 | 9                 | 22.5           |
| Triple negative breast cancer                  | 17                | 42.5           |
| Treatment for breast cancer                    |                   |                |
| Surgery                                        | 31                | 77.5           |
| Systemic chemotherapy                          | 29                | 72.5           |
| Radiotherapy                                   | 8                 | 20.0           |
| NA                                             | 5                 | 12.5           |
| Tumour grade                                   |                   |                |
| Grade 2                                        | 3                 | 7.5            |
| Grade 3                                        | 37                | 92.5           |
| Ki-67 Index (percent)                          |                   |                |
| mean $\pm$ SD                                  | 59.15 $\pm$ 16.88 |                |
| median (IQR)                                   | 60 (50–70)        |                |
| Lymph node status                              |                   |                |
| N0                                             | 7                 | 17.5           |
| N1                                             | 10                | 25.0           |
| N2                                             | 13                | 32.5           |
| N3                                             | 6                 | 15.0           |
| NA                                             | 4                 | 10.0           |
| Extracranial metastasis                        | 28                | 70.0           |
| Pattern of brain metastases                    |                   |                |
| Single                                         | 25                | 62.5           |
| Multiple                                       | 15                | 37.5           |
| Chemotherapy regimen for primary breast cancer |                   |                |
| Anthracycline followed by taxane               | 29                | 72.5           |
| Anti-HER2 regimen                              |                   |                |
| Herceptin                                      | 9                 | 22.5           |
| Treatment of brain metastases                  |                   |                |
| Surgery + WBRT                                 | 5                 | 12.5           |
| Surgery                                        | 2                 | 5.0            |
| WBRT                                           | 7                 | 17.5           |
| No treatment                                   | 26                | 65.0           |
| Death within 6 months                          | 29                | 70.7           |
| Survival time (months)                         |                   |                |
| mean $\pm$ SD                                  | 7.0 $\pm$ 6.11    |                |
| median (IQR)                                   | 4 (3–11.5)        |                |

NA: not available, WBRT: whole brain radiation.

chemotherapy and radiotherapy use between the two groups ( $p = 0.270$  and  $p = 1.00$ ). Tumor grade was mostly grade 3 for both groups ( $p = 1.00$ ), and lymph node status also showed no significant difference ( $p = 0.612$ ).

Extracranial metastases were more common in the HER2+/TNBC group than in the HR + group (84.6% vs. 42.9%,  $p = 0.011$ ), suggesting a higher systemic metastatic spread in HER2+/TNBC patients. However, the pattern of brain metastases (single vs. multiple) was not significantly different between the groups ( $p = 0.392$ ). The anthracycline followed by taxane regimen was the most commonly used chemotherapy in both, with no significant difference ( $p = 0.270$ ). Treatment strategies after the diagnosis of brain metastases were also similar ( $p = 0.944$ ), with no further treatment being the most frequent approach in both subtypes.

### 3.3. Overall survival analysis

A total of 40 patients with breast cancer and brain metastases were included in the survival analysis. During the follow-up period, 29 deaths were observed. The median OS for the entire cohort was 4.0 months (95% CI: 3.23–4.77). The 6-month and 12-month OS rates were 27.5% and 17.5%, respectively (Table 3, Fig. 1).

When stratified by breast cancer subtypes, patients with HR + disease had the longest median OS of 6.0 months (95% CI: 3.58–8.42), followed by HER2+ at 4.0 months (95% CI: 2.34–5.66), and TNBC at 3.0 months (95% CI: 2.34–3.66). No statistically significant difference in OS was found among the three subtypes (log-rank  $p = 0.435$ ) (Table 3, Fig. 2).

No significant differences in OS were found based on the presence of extracranial metastases or the number of brain lesions, with a median OS of 4.0 months (Figs. 3 and 4). However, treatment modality significantly impacted survival ( $p = 0.006$ ). Patients who underwent surgical resection followed by WBRT had the best outcomes, with 6-month and 12-month OS rates of 80.0% and 60.0%, respectively. Those treated with surgery or WBRT alone showed intermediate results, while untreated patients faced the poorest prognosis, with a median OS of 3.0 months and a 12-month OS rate of only 7.7% (Fig. 5).

In univariate analysis, age was a significant factor in overall survival, with those over 60 at higher risk of mortality than those under 50 (HR 3.47;  $p = 0.010$ ). Extracranial metastases and the number of brain metastases did not significantly affect survival. There was a trend toward increased mortality in HER2+ and TNBC breast cancer subtypes compared to HR+, but it was not statistically significant. Patients who received no treatment after brain metastasis diagnosis had a significantly higher risk of death than those treated with surgery plus WBRT (HR 8.54;  $p = 0.036$ ) (Table 4).

In multivariate Cox regression analysis, only age over 60 years was an independent prognostic factor for overall survival (HR 4.38;  $p = 0.041$ ). Other factors, including breast cancer subtypes, extracranial metastases, pattern of brain metastases, and treatment of brain metastases, did not show an independent association with survival (Table 5).

## 4. Discussion

Brain metastases in breast cancer are associated with worse survival rates and an extremely poor prognosis<sup>11–13</sup>. Older age, triple-negative subtype, presence of extracranial metastases, and multiple brain lesions were associated with poorer survival.<sup>11</sup> This study found that approximately 18% of patients with breast cancer developed brain metastases, which aligns with the average incidence reported in various studies (15–30%).<sup>14,15</sup> Additionally, we observed that more than half of the patients diagnosed with brain metastases belonged to the TNBC and HER2+ groups. These findings are consistent with recent studies indicating that the HER2-enriched subtype and TNBC are significant risk factors for brain metastases compared to the HR + subtype.<sup>16,17</sup>

Furthermore, our research highlights several important patterns related to brain metastasis based on breast cancer subtypes. Contrary to many earlier reports that predominantly identified HER2+ or TNBC subtypes among patients with brain metastases, our cohort revealed that 30% of affected patients had HR + disease. This higher percentage may contribute to the comparatively older median age at which brain metastases were diagnosed in our study, approximately 60 years, which is notably older than the commonly reported range of 45–50 years in the existing literature.<sup>18,19</sup>

Consistent with the known biological aggressiveness of HER2+

**Table 2**

Baseline clinicopathological characteristics stratified by breast cancer subtypes.

| Characteristics                                        | HR+ (n = 14)<br>n (%) | HER2+/TNBC (n = 26)<br>n (%) | P-value             |
|--------------------------------------------------------|-----------------------|------------------------------|---------------------|
| <b>Age (years)</b>                                     |                       |                              | 0.770 <sup>b</sup>  |
| 50 or below                                            | 3 (21.4)              | 8 (30.8)                     |                     |
| 51-60                                                  | 5 (35.7)              | 7 (26.9)                     |                     |
| Above 60                                               | 6 (42.9)              | 11 (42.3)                    |                     |
| <b>BMI</b>                                             |                       |                              | 0.753 <sup>b</sup>  |
| 18.5 or below                                          | 1 (7.1)               | 4 (15.4)                     |                     |
| 18.5-22.9                                              | 6 (42.9)              | 10 (38.5)                    |                     |
| Above 23.0                                             | 7 (50.0)              | 12 (46.2)                    |                     |
| <b>TTBM (months), mean <math>\pm</math> SD</b>         | 27.29 $\pm$ 10.07     | 18.65 $\pm$ 6.08             | 0.011 <sup>*a</sup> |
| <b>Treatment for breast cancer</b>                     |                       |                              |                     |
| Surgery                                                | 14 (100)              | 17 (65.4)                    | 0.016 <sup>*c</sup> |
| Systemic chemotherapy                                  | 12 (85.7)             | 17 (65.4)                    | 0.270 <sup>c</sup>  |
| Radiation                                              | 3 (21.4)              | 5 (19.2)                     | 1.00 <sup>c</sup>   |
| <b>Tumour Grade</b>                                    |                       |                              | 1.00 <sup>c</sup>   |
| Grade 2                                                | 1 (7.1)               | 2 (7.7)                      |                     |
| Grade 3                                                | 13 (92.9)             | 24 (92.3)                    |                     |
| <b>Ki-67 Index, mean <math>\pm</math> SD (percent)</b> | 51.14 $\pm$ 13.44     | 63.46 $\pm$ 17.19            | 0.026 <sup>*a</sup> |
| <b>Lymph node status</b>                               |                       |                              | 0.612 <sup>b</sup>  |
| N0                                                     | 3 (21.4)              | 4 (18.2)                     |                     |
| N1                                                     | 5 (35.7)              | 5 (22.7)                     |                     |
| N2                                                     | 5 (35.7)              | 8 (36.4)                     |                     |
| N3                                                     | 1 (7.1)               | 5 (22.7)                     |                     |
| <b>Presence of extracranial metastases</b>             | 6 (42.9)              | 22 (84.6)                    | 0.011 <sup>*c</sup> |
| <b>Pattern of brain metastases</b>                     |                       |                              | 0.392 <sup>b</sup>  |
| Single                                                 | 10 (71.4)             | 15 (57.7)                    |                     |
| Multiple                                               | 4 (28.6)              | 11 (42.3)                    |                     |
| <b>Chemotherapy regimen for primary breast cancer</b>  |                       |                              |                     |
| Anthracycline followed by taxane                       | 12 (85.7)             | 17 (65.4)                    | 0.270 <sup>b</sup>  |
| <b>Treatment of brain metastases</b>                   |                       |                              | 0.944 <sup>b</sup>  |
| Surgery + WBRT                                         | 2 (14.3)              | 3 (11.5)                     |                     |
| Surgery                                                | 1 (7.1)               | 1 (3.8)                      |                     |
| WBRT                                                   | 2 (14.3)              | 5 (19.2)                     |                     |
| No treatment                                           | 9 (64.3)              | 17 (65.4)                    |                     |

TTBM: Time to diagnosis of brain metastases, HR+: Hormone receptor-positive, HER2+: HER2-positive, TNBC: Triple negative breast cancer.

\*Statistically significant at  $\alpha < 0.05$ , statistical use.<sup>a</sup> = Independent *t*-test.<sup>b</sup> = Chi-square test.<sup>c</sup> = Fisher exact test.**Table 3**

Overall survival according to breast cancer subtype, extracranial metastases, pattern of brain metastases, and treatment.

|                                            | Number of deaths | Median OS (months), (95% CI) | OS Rate |          | P-value |
|--------------------------------------------|------------------|------------------------------|---------|----------|---------|
|                                            |                  |                              | 6-month | 12-month |         |
| <b>Overall survival (n = 40)</b>           | 29               | 4.0 (3.23-4.77)              | 27.50%  | 17.50%   |         |
| <b>Breast Cancer Subtypes</b>              |                  |                              |         |          | 0.435   |
| HR+ (n = 14)                               | 8                | 6.0 (3.58-8.42)              | 42.90%  | 21.40%   |         |
| HER2+ (n = 9)                              | 7                | 4.0 (2.34-5.66)              | 22.20%  | 11.10%   |         |
| TNBC (n = 17)                              | 14               | 3.0 (2.34-3.66)              | 17.60%  | 17.60%   |         |
| <b>Presence of extracranial metastases</b> |                  |                              |         |          | 0.921   |
| No (n = 12)                                | 9                | 4.0 (1.74-6.26)              | 50.00%  | 16.70%   |         |
| Yes (n = 28)                               | 20               | 4.0 (2.97-5.03)              | 28.60%  | 17.90%   |         |
| <b>Pattern of brain metastases</b>         |                  |                              |         |          | 0.675   |
| Single (n = 25)                            | 18               | 4.0 (3.19-4.81)              | 28.00%  | 20.00%   |         |
| Multiple (n = 15)                          | 11               | 4.0 (1.73-6.27)              | 26.70%  | 13.30%   |         |
| <b>Treatment of brain metastases</b>       |                  |                              |         |          | 0.006*  |
| Surgery + WBRT (n = 5)                     | 1                | -                            | 80.00%  | 60.00%   |         |
| Surgery (n = 2)                            | 1                | 6.0                          | 50.00%  | 50.00%   |         |
| WBRT (n = 7)                               | 4                | 6.0 (4.29-7.71)              | 42.90%  | 14.30%   |         |
| No treatment (n = 26)                      | 23               | 3.0 (2.23-3.77)              | 11.50%  | 7.70%    |         |

WBRT: whole brain radiation, HR+: Hormone receptor-positive, HER2+: HER2-positive, TNBC: Triple negative breast cancer, OS: overall survival.

\*Statistically significant at  $\alpha < 0.05$ , statistical use: Log-rank test.

and TNBC subtypes, our findings revealed significantly higher rates of extracranial metastases in the HER2+/TNBC group compared to the HR + group (84.6% vs. 42.9%,  $p = 0.011$ ). Additionally, the HER2+/TNBC group displayed a significantly higher Ki-67 index (63.46  $\pm$  17.19 vs. 51.14  $\pm$  13.44,  $p = 0.026$ ), indicating rapid cell

proliferation and aggressive behaviour typical of these subtypes. These results are consistent with existing evidence that suggests HER2 overexpression and TNBC are linked to increased cellular motility, invasiveness, and varied treatment responses, all contributing to a higher risk of widespread metastatic spread.<sup>20,21</sup>

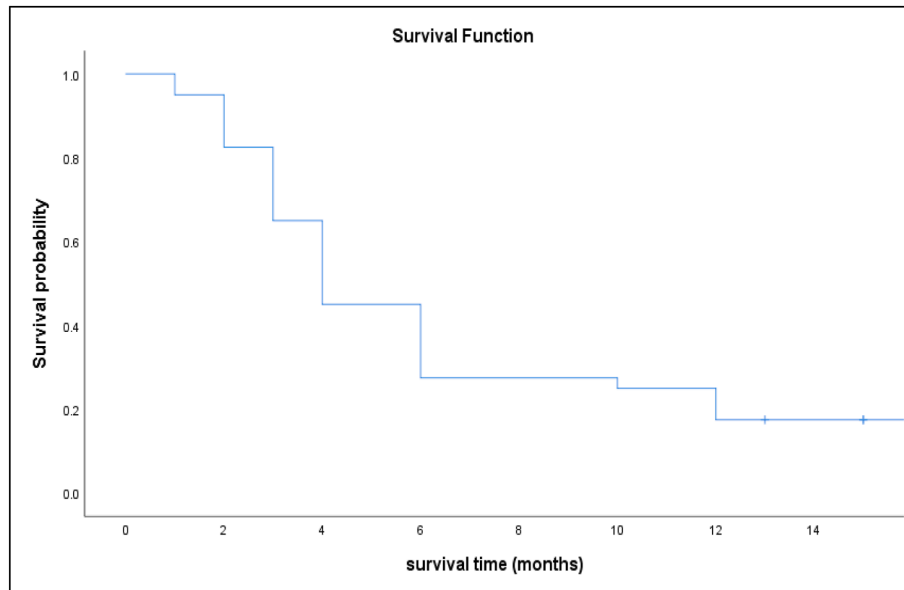


Fig. 1. Kaplan-Meier overall survival curve of breast cancer patients with brain metastases.

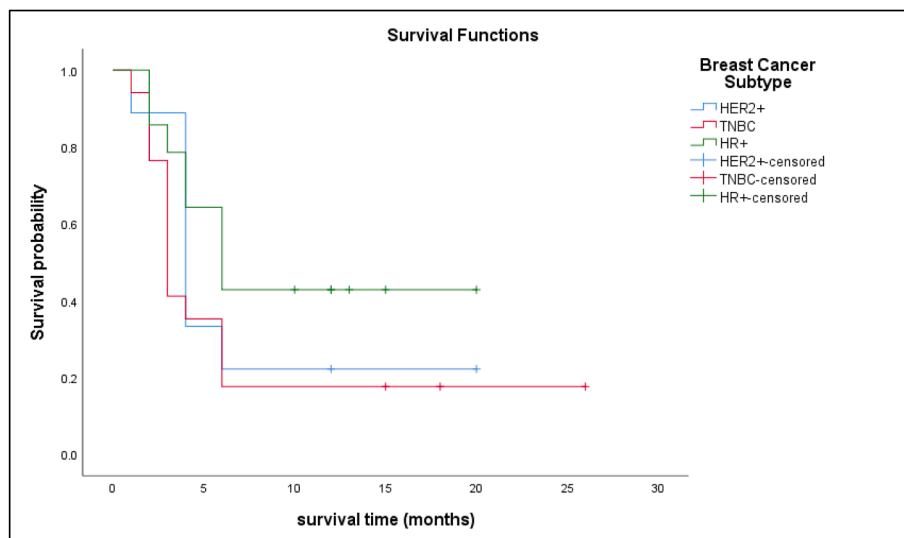


Fig. 2. Kaplan-Meier overall survival curves stratified by breast cancer subtypes.

Our study shows that patients with HR + disease have a significantly longer interval from initial cancer diagnosis to brain metastases compared to those with HER2+/TNBC ( $27.29 \pm 10.07$  vs.  $18.65 \pm 6.08$  months,  $p = 0.011$ ). This aligns with previous reports indicating faster metastatic progression in TNBC.<sup>22,23</sup> The slower disease trajectory and better treatment responses in HR + breast cancer likely explain the longer latency before brain involvement. HER2+ and TNBC tumors exhibit higher proliferation and aggressive metastasis, while HR + tumors are less proliferative and more hormonally responsive, leading to slower progression to central nervous system involvement.

The median OS of 4.0 months in our cohort highlights the poor prognosis of brain metastases in breast cancer, consistent with previous studies reporting OS from 4 to 8 months.<sup>24,25</sup> Breast cancer subtypes showed a trend favouring HR + patients with longer median OS compared to HER2+ and TNBC, although this did not reach statistical significance, the observed trend is consistent

with prior reports indicating more indolent tumor biology and slower disease progression in HR + tumors.<sup>26</sup> In contrast, TNBC has been consistently associated with earlier development of brain metastases and poorer survival outcomes, likely due to higher proliferative activity and limited targeted treatment options. The lack of statistical significance in our cohort is likely attributable to the small sample size and limited power after subtype stratification. Patients with HR + breast cancer generally face a lower risk of developing brain metastases. However, the increasing incidence of resistance to endocrine therapy raises this risk over time.<sup>7,8</sup> In contrast to patients with HER2+ disease, those with HR + cancer and brain metastases have fewer effective systemic treatment options, which may negatively impact their survival and treatment outcomes. Recent therapeutic advancements for HER2+ patients may further improve their prognosis. Trastuzumab deruxtecan (T-DXd), a third-generation antibody-drug conjugate, links trastuzumab to a topoisomerase I inhibitor via a cleavable tetrapeptide

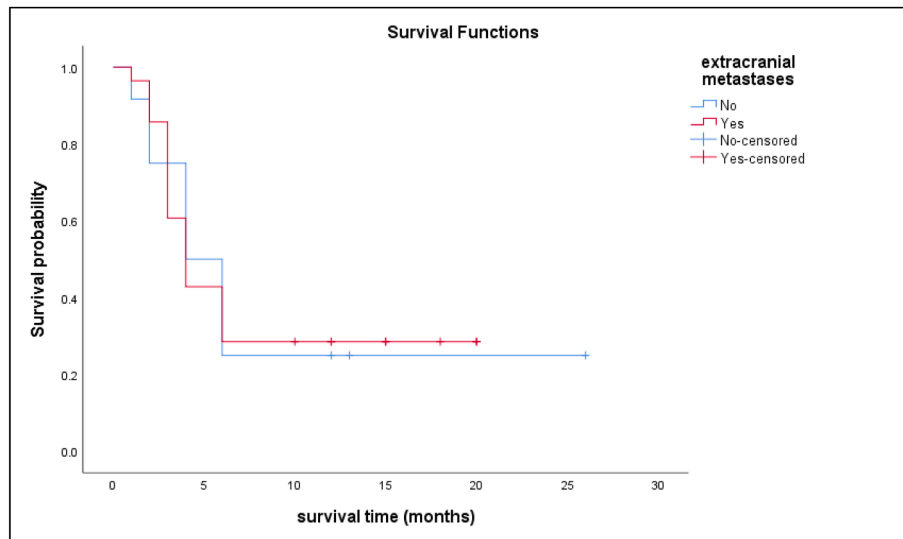


Fig. 3. Kaplan-Meier overall survival curves according to the presence of extracranial metastases.

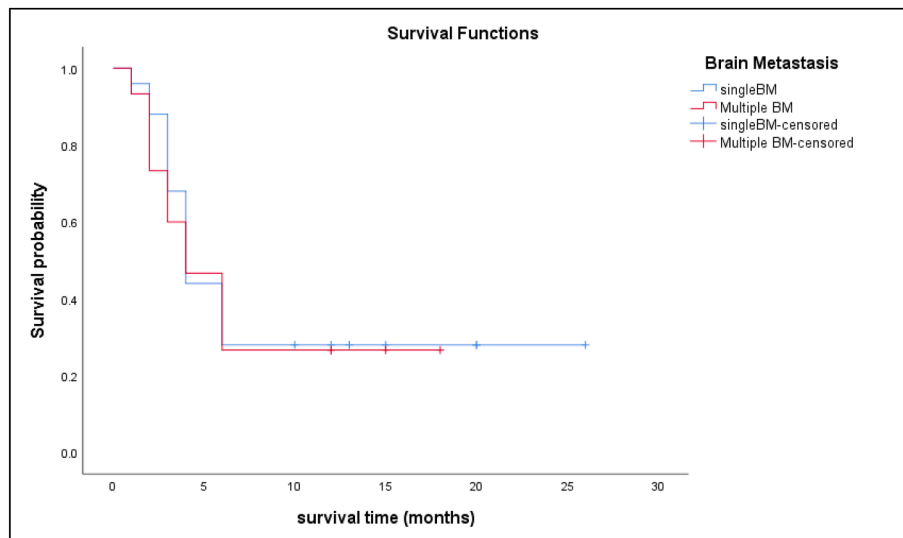


Fig. 4. Kaplan-Meier overall survival curves according to the pattern of brain metastases.

linker and has shown promising activity in the brain. A pooled exploratory analysis from the DESTINY-Breast 01, 02, and 03 trials indicated significant improvements in both the intracranial objective response rate and central nervous system (CNS) progression-free survival for patients with HER2+ metastatic breast cancer who had active or stable brain metastases<sup>11,27–30</sup>. Such emerging treatments may lead to improved survival outcomes for HER2+ patients, a benefit that is currently lacking for those with HR + or TNBC who develop brain metastases.

The treatment modality for brain metastases was shown to influence survival outcomes in univariate analysis, where patients receiving combined surgical resection and WBRT exhibited the most favourable results. This finding aligns with previous studies indicating improved survival rates in selected patients who receive aggressive local therapy, particularly among those with good performance status, limited intracranial disease burden, and controlled extracranial disease.<sup>30</sup> However, after adjusting for confounding factors, the association between treatment modality and survival no longer reached statistical significance, suggesting

potential selection bias and underscoring the impact of patient-related factors in treatment decision-making.

In multivariate analysis, age emerged as the only independent prognostic factor, with patients older than 60 years demonstrating significantly worse survival outcomes. This finding is consistent with established prognostic frameworks, such as the Graded Prognostic Assessment (GPA), which identifies age as a key determinant of survival in patients with brain metastases.<sup>31</sup> Older age may be associated with reduced treatment tolerance, a higher burden of comorbidities, and decreased eligibility for aggressive multimodal therapy, all of which may contribute to poorer outcomes.

Interestingly, neither the presence of extracranial metastases nor the quantity of brain metastatic lesions showed a significant association with survival in this cohort. Although these factors have been deemed prognostic in larger studies,<sup>31,32</sup> their influence may be diminished in small cohorts characterized by uniformly advanced disease and brief survival periods. This underscores the heterogeneity of breast cancer brain metastases and emphasizes



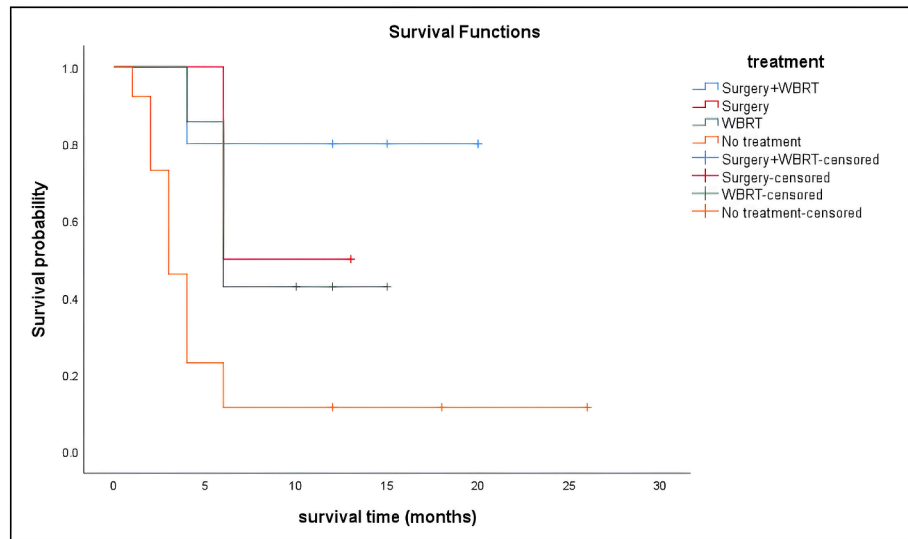


Fig. 5. Kaplan-Meier overall survival curves according to treatment of brain metastases.

Table 4

Univariate Cox regression analysis for overall survival.

| Variables                                  | HR   | 95% CI    |       | P-value |
|--------------------------------------------|------|-----------|-------|---------|
|                                            |      | Lower     | Upper |         |
| <b>Age (years)</b>                         |      |           |       |         |
| 50 or below                                | 1.00 | reference |       |         |
| 51-60                                      | 1.39 | 0.46      | 4.15  | 0.557   |
| Above 60                                   | 3.47 | 1.34      | 9.01  | 0.010*  |
| <b>Breast Cancer Subtypes</b>              |      |           |       |         |
| HR+                                        | 1.00 | reference |       |         |
| HER2+                                      | 1.54 | 0.56      | 4.27  | 0.407   |
| TNBC                                       | 2.08 | 0.87      | 4.97  | 0.102   |
| <b>Presence of extracranial metastases</b> |      |           |       |         |
| No                                         | 1.00 | reference |       |         |
| Yes                                        | 1.01 | 0.46      | 2.22  | 0.981   |
| <b>Pattern of brain metastases</b>         |      |           |       |         |
| Single                                     | 1.00 | reference |       |         |
| Multiple                                   | 1.09 | 0.515     | 2.31  | 0.821   |
| <b>Treatment of brain metastases</b>       |      |           |       |         |
| Surgery + WBRT                             | 1.00 | reference |       |         |
| Surgery                                    | 2.29 | 0.14      | 36.63 | 0.558   |
| WBRT                                       | 2.78 | 0.31      | 24.91 | 0.360   |
| No treatment                               | 8.54 | 1.15      | 63.63 | 0.036*  |

WBRT: whole brain radiation, HR+: Hormone receptor positive.

HER2+: HER2-positive, TNBC: Triple negative breast cancer, HR: Hazard ratio.

\*Statistically significant at  $\alpha < 0.05$ .

Table 5

Multivariate Cox regression analysis for overall survival.

| Variables                                  | HR   | 95% CI    |       | P-value |
|--------------------------------------------|------|-----------|-------|---------|
|                                            |      | Lower     | Upper |         |
| <b>Age (years)</b>                         |      |           |       |         |
| 50 or below                                | 1.00 | reference |       |         |
| 51-60                                      | 1.50 | 0.38      | 5.84  | 0.562   |
| Above 60                                   | 4.38 | 1.06      | 18.11 | 0.041*  |
| <b>Breast Cancer Subtypes</b>              |      |           |       |         |
| HR+                                        | 1.00 | reference |       |         |
| HER2 +                                     | 2.38 | 0.66      | 8.63  | 0.187   |
| TNBC                                       | 2.33 | 0.72      | 7.54  | 0.157   |
| <b>Presence of extracranial metastases</b> |      |           |       |         |
| No                                         | 1.00 | reference |       |         |
| Yes                                        | 2.67 | 0.81      | 8.81  | 0.108   |
| <b>Pattern of brain metastases</b>         |      |           |       |         |
| Single                                     | 1.00 | reference |       |         |
| Multiple                                   | 1.08 | 0.46      | 2.55  | 0.860   |
| <b>Treatment of brain metastases</b>       |      |           |       |         |
| Surgery + WBRT                             | 1.00 | reference |       |         |
| Surgery                                    | 3.28 | 0.19      | 55.38 | 0.411   |
| WBRT                                       | 2.50 | 0.24      | 25.68 | 0.442   |
| No treatment                               | 7.10 | 0.80      | 62.81 | 0.078   |

WBRT: whole brain radiation, HR+: Hormone receptor positive.

HER2+: HER2-positive, TNBC: Triple-negative breast cancer, HR: Hazard ratio.

\*Statistically significant at  $\alpha < 0.05$ .

the necessity for tailored prognostic assessments.

This study has several limitations. The retrospective nature, small sample size, and data were collected from a single centre limit the generalizability of the findings and preclude definitive conclusions regarding independent prognostic factors. Future studies with larger, multi-institutional cohorts are needed to document these findings.

## 5. Conclusion

Overall survival for patients with breast cancer brain metastases is still relatively poor. Advanced age has been identified as an independent predictor of poorer survival outcomes. In contrast, aggressive brain-directed treatments, especially when combining surgery with WBRT, have been linked to improved results. These findings underscore the necessity for personalized prognostic

assessments and multidisciplinary treatment approaches to enhance care for this high-risk patient population.

## Conflict of interest

We have no known conflict of interest to disclose.

## References

- Lakha F, Suriyawongpaisul P, Sangrajrang S, Leerapan B, Coker R. Breast cancer in Thailand: policy and health system challenges to universal healthcare. *Health Pol Plann*. 2020 Nov 20;35(9):1159–1167. <https://doi.org/10.1093/heapol/czaa063>.
- Raghavendra AS, Ibrahim NK. Breast cancer brain metastasis: a comprehensive review. *JCO Oncol Pract*. 2024 Oct;20(10):1348–1359. <https://doi.org/10.1200/OP.23.00794>.
- Chen Q, Xiong J, Ma Y, Wei J, Liu C, Zhao Y. Systemic treatments for breast cancer brain metastasis. *Front Oncol*. 2023 Jan 6;12:1086821. <https://doi.org/10.3389/fonc.2022.1086821>.

4. Bailleux C, Eberst L, Bachelot T. Treatment strategies for breast cancer brain metastases. *Br J Cancer*. 2021 Jan;124(1):142–155. <https://doi.org/10.1038/s41416-020-01175-y>.
5. Evans AJ, James JJ, Cornford EJ, et al. Brain metastases from breast cancer: identification of a high-risk group. *Clin Oncol (R Coll Radiol)*. 2004 Aug;16(5):345–349. <https://doi.org/10.1016/j.clon.2004.03.012>.
6. Chow L, Suen D, Ma KK, Kwong A. Identifying risk factors for brain metastasis in breast cancer patients: implication for a vigorous surveillance program. *Asian J Surg*. 2015 Oct;38(4):220–223. <https://doi.org/10.1016/j.asjsur.2015.03.003>.
7. Willman M, Willman J, Lucke-Wold B. Endocrine resistant breast cancer: brain metastasis. *Explor Target Antitumor Ther*. 2022;3(2):240–251. <https://doi.org/10.37349/etat.2022.00081>.
8. Hartkopf AD, Grischke EM, Brucker SY. Endocrine-resistant breast cancer: mechanisms and treatment. *Breast Care (Basel)*. 2020 Aug;15(4):347–354. <https://doi.org/10.1159/000508675>.
9. Seoane J, De Mattos-Arruda L. Brain metastasis: new opportunities to tackle therapeutic resistance. *Mol Oncol*. 2014 Sep 12;8(6):1120–1131. <https://doi.org/10.1016/j.molonc.2014.05.009>.
10. Olson EM, Abdel-Rasoul M, Maly J, Wu CS, Lin NU, Shapiro CL. Incidence and risk of central nervous system metastases as site of first recurrence in patients with HER2-positive breast cancer treated with adjuvant trastuzumab. *Ann Oncol*. 2013 Jun;24(6):1526–1533. <https://doi.org/10.1093/annonc/mdt036>.
11. Durrani S, Al-Ghamdi AA, Al-Bugawi A, et al. Prognostic factors and survival outcome of brain metastases in breast cancer patients: a retrospective analysis. *BMC Cancer*. 2025 Sep 30;25(1):1455. <https://doi.org/10.1186/s12885-025-14844-z>.
12. Bendrich S, Pietrkiewicz MAW, Schirmer MA, et al. Borderline decisions in brain-metastatic breast cancer: efficacy of multimodality treatments in breast cancer patients with very limited prognosis suffering from brain metastases. *Clin Exp Metastasis*. 2025 Oct 27;42(6):62. <https://doi.org/10.1007/s10585-025-10376-9>.
13. Riecke K, Müller V, Neunhöfner T, et al. Long-term survival of breast cancer patients with brain metastases: subanalysis of the BMBC registry. *ESMO Open*. 2023 Jun;8(3):101213. <https://doi.org/10.1016/j.esmoop.2023.101213>.
14. Tabouret E, Chinot O, Metellus P, Tallet A, Viens P, Gonçalves A. Recent trends in epidemiology of brain metastases: an overview. *Anticancer Res*. 2012 Nov;32(11):4655–4662.
15. Leone JP, Leone BA. Breast cancer brain metastases: the last frontier. *Exp Hematol Oncol*. 2015 Nov 24;4:33. <https://doi.org/10.1186/s40164-015-0028-8>.
16. Sun MS, Yun YY, Liu HJ, Yu ZH, Yang F, Xu L. Brain metastasis in de novo breast cancer: an updated population-level study from SEER database. *Asian J Surg*. 2022 Nov;45(11):2259–2267. <https://doi.org/10.1016/j.asjsur.2021.12.037>.
17. Zimmer AS. Triple-negative breast cancer central nervous system metastases from the laboratory to the clinic. *Cancer J*. 2021 Jan-Feb 01;27(1):76–82. <https://doi.org/10.1097/PPO.0000000000000503>.
18. Lentzsch S, Reichardt P, Weber F, Budach V, Dörken B. Brain metastases in breast cancer: prognostic factors and management. *Eur J Cancer*. 1999 Apr;35(4):580–585. [https://doi.org/10.1016/s0959-8049\(98\)00421-3](https://doi.org/10.1016/s0959-8049(98)00421-3).
19. Bottosso M, Griguolo G, Guiu S, et al. Temporal evolution of breast cancer brain metastases treatments and outcomes. *npj Breast Cancer*. 2025 Feb 20;11(1):20. <https://doi.org/10.1038/s41523-025-00735-w>.
20. Maqsood Q, Khan MU, Fatima T, Khalid S, Malik ZI. Recent insights into breast cancer: molecular pathways, epigenetic regulation, and emerging targeted therapies. *Breast Cancer (Auckl)*. 2025 Jul 13;19:11782234251355663. <https://doi.org/10.1177/11782234251355663>.
21. Xu D, Hu Z, Wang K, et al. Why does HER2-positive breast cancer metastasize to the brain and what can we do about it? *Crit Rev Oncol Hematol*. 2024 Mar;195:104269. <https://doi.org/10.1016/j.critrevonc.2024.104269>. Epub 2024 Jan 23. PMID: 38272149.
22. Niwińska A, Murawska M, Pogoda K. Breast cancer brain metastases: differences in survival depending on biological subtype, RPA RTOG prognostic class and systemic treatment after whole-brain radiotherapy (WBRT). *Ann Oncol*. 2010 May;21(5):942–948. <https://doi.org/10.1093/annonc/mdp407>.
23. Bustamante E, Casas F, Luque R, et al. Brain metastasis in triple-negative breast cancer. *Breast J*. 2024 Sep 30;2024:8816102. <https://doi.org/10.1155/2024/8816102>.
24. Niwińska A, Murawska M, Pogoda K. Breast cancer brain metastases: differences in survival depending on biological subtype and treatment. *J Neuro Oncol*. 2010;96(2):281–288.
25. Dawood S, Gonzalez-Angulo AM, Albarracín C, et al. Brain metastases as the first site of recurrence in patients with breast cancer. *Clin Breast Cancer*. 2012;12(2):85–91.
26. Kennecke H, Yerushalmi R, Woods R, et al. Metastatic behavior of breast cancer subtypes. *J Clin Oncol*. 2010;28(20):3271–3277.
27. Hurvitz SA, Modi S, Li W, et al. 3770 A pooled analysis of trastuzumab deruxtecan (T-DXd) in patients (pts) with HER2-positive (HER2<sup>+</sup>) metastatic breast cancer (mBC) with brain metastases (BMs) from DESTINY-Breast (DB)-01, -02, and -03. *Ann Oncol*. 2023;34:S335–S336.
28. Modi S, Jacot W, Yamashita T, et al. Trastuzumab deruxtecan in previously treated her2-low advanced breast cancer. *N Engl J Med*. 2022 Jul 7;387(1):9–20. <https://doi.org/10.1056/NEJMoa2203690>.
29. Mosele F, Deluche E, Lusque A, et al. Trastuzumab deruxtecan in metastatic breast cancer with variable HER2 expression: the phase 2 DAISY trial. *Nat Med*. 2023 Aug;29(8):2110–2120. <https://doi.org/10.1038/s41591-023-02478-2>.
30. Harbeck N, Ciruelos E, Jerusalem G, et al. Trastuzumab deruxtecan in HER2-positive advanced breast cancer with or without brain metastases: a phase 3b/4 trial. *Nat Med*. 2024 Dec;30(12):3717–3727. <https://doi.org/10.1038/s41591-024-03261-7>.
31. Sperduto PW, Mesko S, Li J, et al. Survival in patients with brain metastases: summary report on the updated diagnosis-specific graded prognostic assessment (DS-GPA). *J Clin Oncol*. 2020;38(32):3773–3784.
32. Nieder C, Mehta MP. Prognostic indices for brain metastases—usefulness and challenges. *Radiat Oncol*. 2009;4:10.